

Scalability & Future Plan

Date	2 July 2026
Team ID	SWTID-2026-7147
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current Smart Lender solution can be scaled to support larger user bases, higher transaction volumes, increasing datasets, and future AI enhancements. It also presents the roadmap for improving the application's scalability, performance, security, and long-term maintainability.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Flask application is deployed on a single server, limiting concurrent users.	Reduced performance during high traffic.	High
2	Dataset is static and relatively small, which may affect prediction accuracy on diverse applicants.	Lower model generalization.	High
3	No real-time monitoring, logging, or model retraining pipeline.	Difficult to maintain long-term accuracy and reliability.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask instance	Deploy multiple instances using Docker and Kubernetes with load balancing.
2	Data Storage	Local dataset	Migrate to a scalable cloud database such as PostgreSQL, IBM Db2, or MongoDB with cloud storage.
3	Performance	Predictions served by one application server	Use caching, asynchronous request processing, optimized XGBoost inference, and auto-scaling.
4	Security	Basic input validation	Implement HTTPS, JWT authentication, role-based access control, encrypted databases, and regular security audits.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Deploy application using Docker containers	1–2 Months	Improved deployment consistency and scalability
Phase 2	Implement Kubernetes with Load Balancer	2–3 Months	Handle thousands of concurrent users efficiently
Phase 3	Integrate Cloud Database and Monitoring Dashboard	3–4 Months	Better reliability, storage scalability, and system monitoring
Phase 4	Continuous Model Retraining and CI/CD Pipeline	4–6 Months	Higher prediction accuracy and automated updates